Lab 3: Frontend, Ingress & Persistent Storage

From a Blank Folder to a Complete 3-Tier Stack Behind One Entry Point

What You Will Build

By the end of this lab you will have created an Angular application from scratch, containerized it, and deployed it to Kubernetes alongside your existing PostgreSQL and .NET Core API from Lab 2. A single Ingress will route browser traffic to either the frontend or the backend based on URL path. PostgreSQL will be backed by a PersistentVolume so its data survives Pod restarts — closing the gap deliberately left open in Lab 1.

Learning Objectives

- Scaffold and build an Angular application for production
- Containerize a static frontend using a multi-stage Docker build with nginx
- Install and understand the role of an Ingress Controller
- Configure path-based routing through a single Ingress resource
- Understand PersistentVolumes, PersistentVolumeClaims, and why Pods alone cannot retain data

Prerequisites

- Node.js and npm installed (node --version, npm --version)
- Angular CLI installed globally: npm install -g @angular/cli
- Lab 2 completed: postgres and api Pods running and reachable via /db-check
- Docker Desktop with Kubernetes enabled

Part A — Create the Angular Application

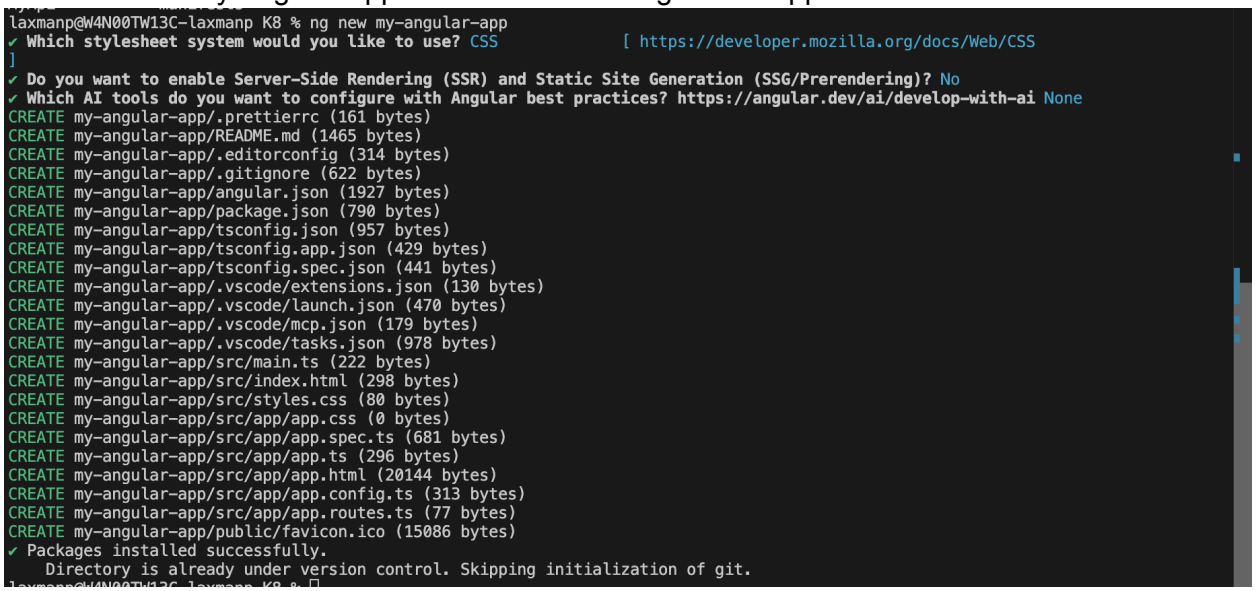
Step A1 — Scaffold a new Angular project

Run this from a clean working directory, outside your MyApi folder.

```
cd C:\k8s-training
ng new my-angular-app
```

When prompted, choose the defaults: no SSR/SSG (Server-Side Rendering) unless you specifically want it — this lab assumes a plain client-side rendered app, which is the simpler and more common starting point. Choose CSS for styling unless you have a preference.

This creates a my-angular-app folder with a working starter application.



Step A2 — Add a simple call to your backend API

Open my-angular-app/src/app/app.ts (or app.component.ts, depending on your Angular CLI version) and replace its contents with the following. This adds a button that calls your API's /db-check endpoint and displays the result, proving the full chain works once deployed.

src/app/app.ts

```
import { Component, signal } from '@angular/core';
import { HttpClient } from '@angular/common/http';

@Component({
  selector: 'app-root',
  standalone: true,
  template: `
    <div style="font-family: sans-serif; padding: 2rem;">
      <h1>Kubernetes 3-Tier Demo</h1>
      <button (click)="checkDb()">Check Database Connection</button>
      <pre>{{ result() }}</pre>
    </div>
  `,
})
export class App {
  result = signal('Click the button to test the API -> Postgres connection');

  constructor(private http: HttpClient) {}

  checkDb() {
    this.http.get('/api/db-check').subscribe({
      next: (res) => this.result.set(JSON.stringify(res, null, 2)),
      error: (err) => this.result.set('Error: ' + JSON.stringify(err.message)),
    });
  }
}
```

Step A3 — Enable HttpClient

Open src/app/app.config.ts and ensure provideHttpClient() is included, so the HttpClient used above actually works.

src/app/app.config.ts

```
import { ApplicationConfig } from '@angular/core';
import { provideHttpClient } from '@angular/common/http';

export const appConfig: ApplicationConfig = {
  providers: [provideHttpClient()],
};
```

Why the URL is /api/db-check, not the full backend address

The Angular app deliberately calls a relative path, not a hostname or port. Once deployed, the Ingress we configure in Part C will intercept any request starting with /api and forward it to the backend Service. The frontend code never needs to know the backend's actual location.

Step A4 — Build for production

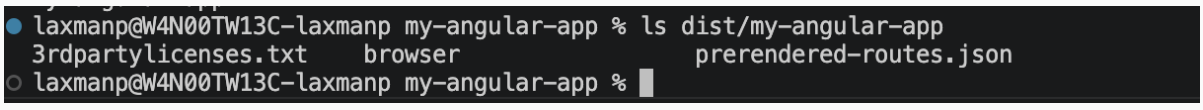
```
cd my-angular-app
ng build
```

Once complete, check the exact output path — this varies by Angular version:

```
dir dist\my-angular-app
```

Important

Angular 17 and later nest build output under a /browser subfolder (dist/my-angular-app/browser) to support server-side rendering setups. Older versions place files directly under dist/my-angular-app. Confirm which structure you have now -- you will need the exact path in Step B1.



Part B — Containerize the Angular App

Step B1 — Create the Dockerfile

This is a multi-stage build, the same pattern used for the .NET API: stage one compiles the app, stage two serves only the compiled static files using nginx. Adjust the COPY path in stage two to match what you found in Step A4.

Dockerfile

```
# Stage 1: Build the Angular app
FROM node:20 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .
RUN npm run build

# Stage 2: Serve with nginx
FROM nginx:alpine
# Adjust this path to match your actual dist output from Step A4
COPY --from=build /app/dist/my-angular-app/browser /usr/share/nginx/html
COPY nginx.conf /etc/nginx/conf.d/default.conf
EXPOSE 80
```

Step B2 — Create nginx.conf

This single setting is the most commonly missed step when containerizing any single-page application. Without it, refreshing the browser on any route other than the homepage returns a 404, because nginx looks for a literal file at that path and finds none.

nginx.conf

```
server {
    listen 80;
    root /usr/share/nginx/html;
    index index.html;

    location / {
        try_files $uri $uri/ /index.html;
    }
}
```

Place both Dockerfile and nginx.conf in the root of my-angular-app, alongside package.json.

Step B3 — Build the image

```
docker build -t myangularapp:latest .
```

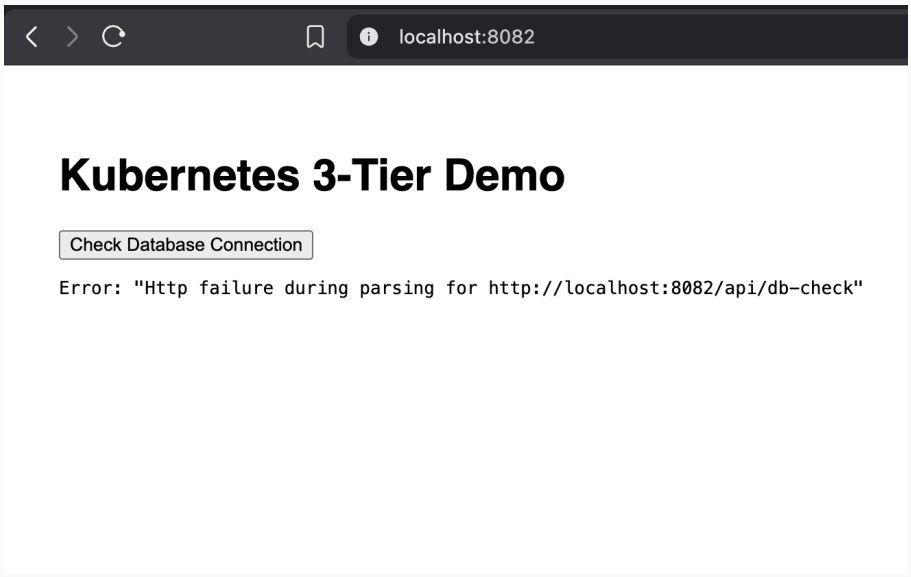
```
 => [stage-1 2/3] COPY --from=build /app/dist/my-angular-app/browser /usr/share/nginx/html
 => [stage-1 3/3] COPY nginx.conf /etc/nginx/conf.d/default.conf
 => exporting to image
 => => exporting layers
 => => exporting manifest sha256:42917203bb300c089df6ba45f3666cc74e7303dce58faa2ac889643525a42e87
 => => exporting config sha256:0c8c4facef936024fcd6e9707af34c250513d80daa92628378537979c504fea0
 => => exporting attestation manifest sha256:f8c5907ed490f9cc673ee687c032b679f9f9b9af2def7e6de19d385e1
 => => exporting manifest list sha256:0655caa2604e3b028a41f1ca334a6fc11f11ff3744c652b053fa01b4562e8095
 => => naming to docker.io/library/myangularapp:latest
 => => unpacking to docker.io/library/myangularapp:latest
laxmanp@W4N00TW13C-laxmanp my-angular-app % docker images

IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
alpine:latest                                  28bd5fe8b56d       13.6MB         4.27MB
compose-postgres-api-lab-api:latest            309a53ba5531        387MB           97MB    U
docker/desktop-cloud-provider-kind:v0.5.0      4ad59ce20658        574MB          150MB    U
docker/desktop-containerd-registry-mirror:v0.0.3   179a9ea3ae47       48.8MB         15.5MB    U
dpage/pgadmin4:latest                          cefc4cc6b7d9        765MB          175MB    U
envoyproxy/envoy:v1.36.4                       08acc53db6f4        233MB         58.8MB
fullstack-ubuntu-web:v1                        34e24cb31c05        2.57GB          595MB
kindest/node:v1.34.3                           08497ee19eac        1.33GB          364MB    U
myangularapp:latest                            0655caa2604e        91.9MB         25.9MB
```

Step B4 — Smoke-test locally

```
docker run -p 8082:80 myangularapp:latest
```

Open http://localhost:8082/ in a browser. You should see the page with the "Check Database Connection" button. Clicking it will fail right now with a network error -- that is expected, since /api/db-check has nothing to route to outside the cluster. Stop the container with Ctrl+C once confirmed the page loads.



Browser showing the Angular app loaded at localhost:8082

Step B5 — Make the image visible to your cluster (kind only)

> **Skip this step if you chose Kubeadm**
>
> This step only applies if Docker Desktop's Kubernetes was set up using the kind engine. If you chose Kubeadm instead, your cluster shares the host's Docker image cache directly, and this step is unnecessary -- continue straight to Part C.

A kind cluster runs its node as a separate container with its own isolated container runtime and image storage, entirely separate from your host machine's Docker image cache. Building an image with docker build on your host does not automatically make it visible inside the kind node. Skipping this step is the most common cause of an ImagePullBackOff or ErrImagePull error in Part C.

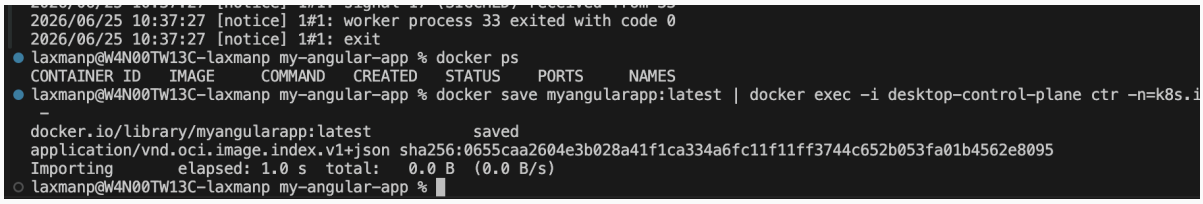
First, find the exact name of your node container:

```
docker ps
```

Look for a container whose name ends in -control-plane (for example, desktop-control-plane). Then load the image directly into that node's containerd runtime:

```
docker save myangularapp:latest | docker exec -i desktop-control-plane ctr -n=k8s.io
images import -
```

Replace desktop-control-plane with your actual node container name if different. Repeat this step any time you rebuild the image -- the kind node never picks up a rebuild automatically.



Part C — Deploy the Frontend to Kubernetes

Step C1 — Create the frontend manifest

03-frontend.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend
  labels:
    app: frontend
spec:
  replicas: 2
  selector:
    matchLabels:
      app: frontend
  template:
    metadata:
      labels:
        app: frontend
    spec:
      containers:
        - name: frontend
          image: myangularapp:latest
          imagePullPolicy: Never
          ports:
            - containerPort: 80
          resources:
            requests:
              cpu: "50m"
              memory: "64Mi"
            limits:
              cpu: "200m"
              memory: "256Mi"
          readinessProbe:
            httpGet:
              path: /
              port: 80
            initialDelaySeconds: 5
            periodSeconds: 5
          livenessProbe:
```

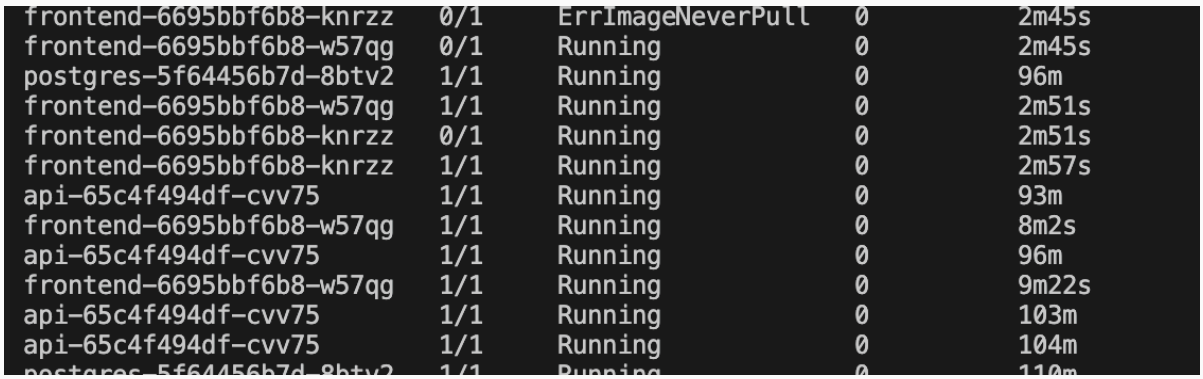
```
            httpGet:
              path: /
              port: 80
            initialDelaySeconds: 10
            periodSeconds: 15
---
apiVersion: v1
kind: Service
metadata:
  name: frontend
spec:
  selector:
    app: frontend
  ports:
    - port: 80
      targetPort: 80
  type: ClusterIP
```

Note the replica count of 2, not 1. Serving static files is cheap and entirely stateless, so there is no reason to run only a single copy from the start.

Step C2 — Deploy it

```
kubectl apply -f 03-frontend.yaml
kubectl get pods --watch
```

Wait until both frontend pods show 1/1 Running, then press Ctrl+C.



Part D — Install an Ingress Controller

Defining an Ingress resource by itself does nothing on its own -- it is only a routing rule. Something must actually read that rule and implement the routing: that something is the Ingress Controller, a real running piece of software. Docker Desktop does not install one by default, so this step installs it explicitly.

Step D1 — Install the nginx Ingress Controller

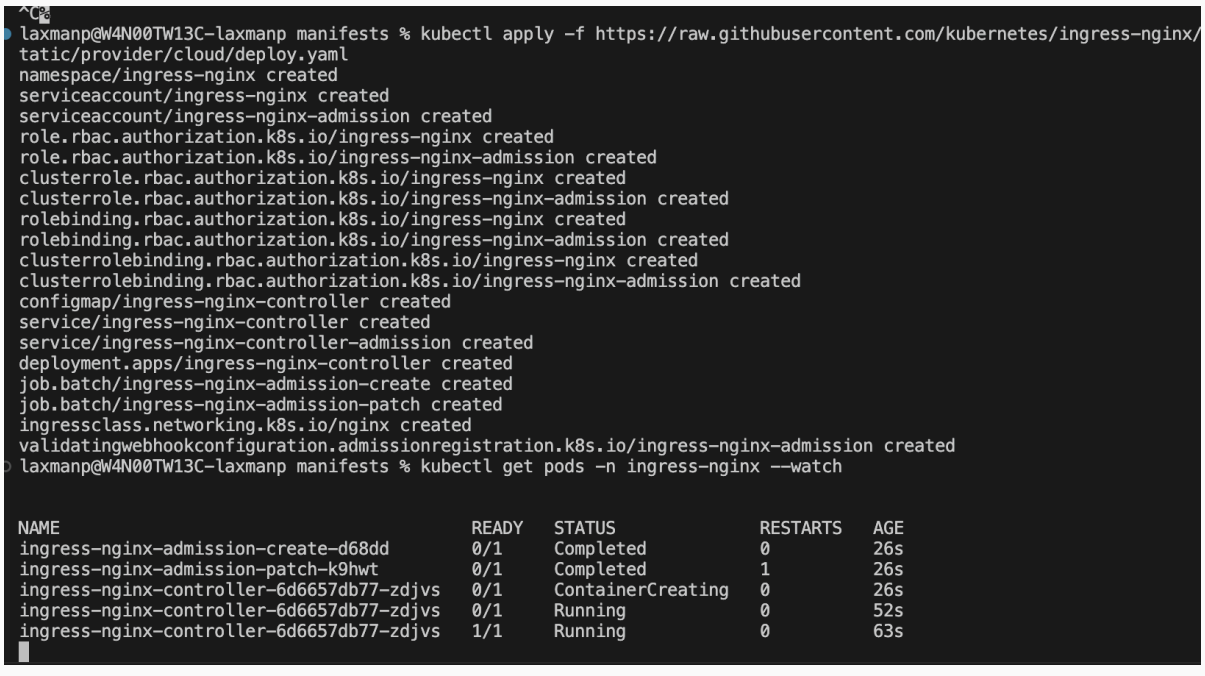
```
kubectl apply -f
https://raw.githubusercontent.com/kubernetes/ingress-nginx/controller-v1.11.2/deploy/
static/provider/cloud/deploy.yaml
```

This deploys the controller into its own namespace, separate from your application Pods.

Step D2 — Wait for the controller to be ready

```
kubectl get pods -n ingress-nginx --watch
```

Wait for a pod named ingress-nginx-controller-... to reach 1/1 Running. This can take 30-60 seconds. Press Ctrl+C once ready.



Part E — Configure the Ingress

A subtle bug worth understanding up front

A single Ingress resource with one rewrite-target annotation covering both the frontend and backend paths looks reasonable, but breaks the frontend in a specific way: annotations apply to the entire Ingress resource, not to individual paths within it. The rewrite-target rule, intended only to strip /api, ends up rewriting every request that flows through that Ingress -- including requests for the Angular app's own JavaScript and CSS files. The browser then receives the wrong content for those files and throws a MIME-type error. The fix is to use two separate Ingress resources, so the rewrite only ever applies to API traffic.

Step E1 — Create the frontend Ingress

This resource has no rewrite annotation at all. Requests pass through to the frontend Service exactly as received.

03-frontend-ingress.yaml

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: frontend-ingress
spec:
  ingressClassName: nginx
  rules:
    - http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: frontend
                port:
                  number: 80
```

Step E2 — Create the API Ingress

This resource keeps the rewrite-target annotation, but it is now scoped to its own Ingress resource and therefore only ever affects API traffic. It strips the /api prefix before forwarding, so the request arrives at the backend exactly as the route definitions expect (e.g. /db-check, not /api/db-check).

03-api-ingress.yaml

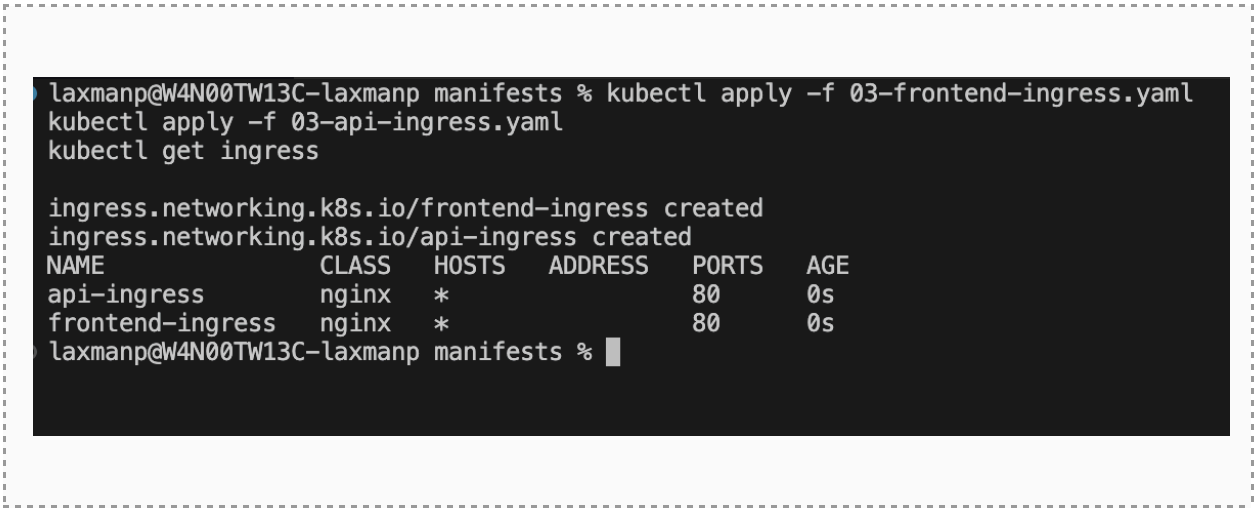
```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: api-ingress
  annotations:
    nginx.ingress.kubernetes.io/rewrite-target: /$2
spec:
```

```
  ingressClassName: nginx
  rules:
    - http:
        paths:
          - path: /api(/|$)(.*)
            pathType: ImplementationSpecific
            backend:
              service:
                name: api
                port:
                  number: 8080
```

Step E3 — Apply and verify both

```
kubectl apply -f 03-frontend-ingress.yaml
kubectl apply -f 03-api-ingress.yaml
kubectl get ingress
```

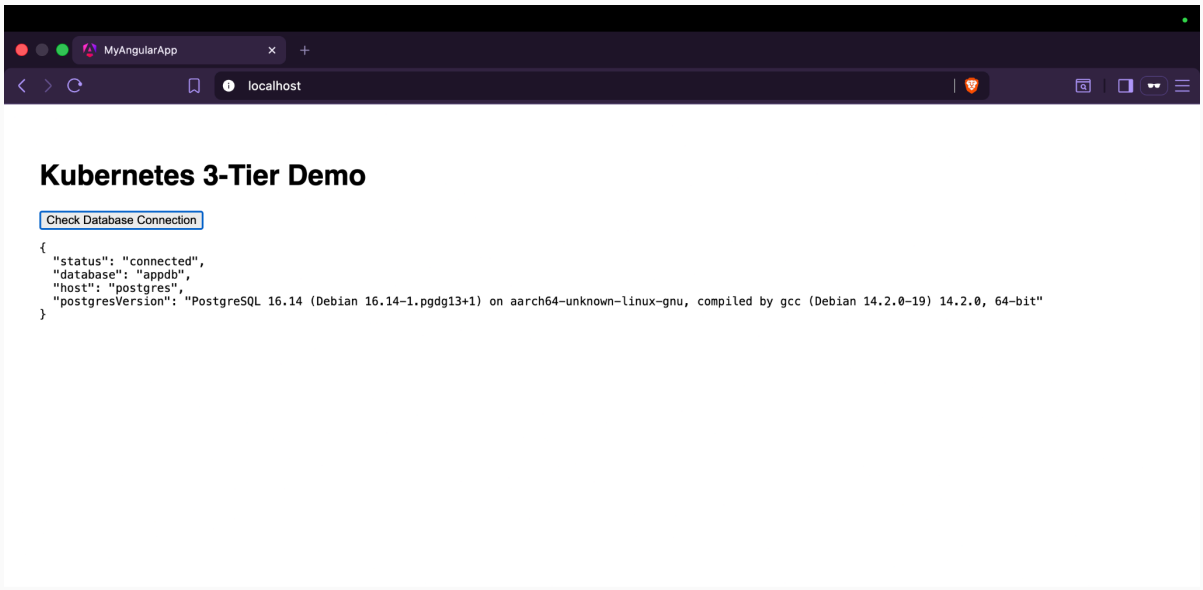
Expect to see two separate Ingress entries listed, frontend-ingress and api-ingress, both showing localhost under ADDRESS.



Step E4 — Test the full stack through one entry point

Use an Incognito/InPrivate window for this test, to rule out any stale cached files from earlier attempts.

1. Open http://localhost/ in a fresh Incognito/InPrivate window -- this should load your Angular app with no console errors.
2. Click the "Check Database Connection" button.
3. Confirm it returns a successful JSON response with the Postgres version, instead of the network error seen in Step B4.



This proves the entire chain end-to-end: browser, to Ingress, to frontend or backend Service by path, to the API Pod, to the Postgres Service by DNS name, to the Postgres Pod.

Part F — Persistent Storage for PostgreSQL

In Lab 1, deleting a Pod proved Kubernetes' self-healing behavior -- but it also proved something uncomfortable: the replacement Pod got a completely empty filesystem. Any data PostgreSQL had written was gone. This part closes that gap.

Step F1 — Understand the two objects involved

PersistentVolume (PV) vs PersistentVolumeClaim (PVC)

A PersistentVolume is a piece of real storage in the cluster -- a chunk of disk space, independent of any particular Pod's lifecycle.

A PersistentVolumeClaim is a request for storage made by a Pod -- "I need 1Gi of space." Kubernetes matches claims to available volumes.

This separation means storage survives even if the Pod requesting it is deleted and recreated. The new Pod's claim binds to the same existing volume, not a new, empty one.

Step F2 — Create the PersistentVolumeClaim

On Docker Desktop, a default StorageClass automatically provisions the underlying volume for you, so only a PVC needs to be defined explicitly.

03-postgres-pvc.yaml

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: postgres-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
```

ReadWriteOnce means the volume can be mounted as read-write by a single node at a time -- the correct mode for a single-instance database like this one.

Step F3 — Update the PostgreSQL Deployment to use it

Two additions to the existing Deployment from Lab 2: a volumeMounts entry inside the container, and a volumes entry at the Pod spec level referencing the PVC.

03-postgres-with-storage.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
```

```
  name: postgres
  labels:
    app: postgres
spec:
  replicas: 1
  selector:
    matchLabels:
      app: postgres
  template:
    metadata:
      labels:
        app: postgres
    spec:
      containers:
        - name: postgres
          image: postgres:16
          ports:
            - containerPort: 5432
          env:
            - name: POSTGRES_DB
              value: appdb
            - name: POSTGRES_USER
              value: appuser
            - name: POSTGRES_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: postgres-secret
                  key: DB_PASSWORD
            - name: PGDATA
              value: /var/lib/postgresql/data/pgdata
          volumeMounts:
            - name: postgres-storage
              mountPath: /var/lib/postgresql/data
      volumes:
        - name: postgres-storage
          persistentVolumeClaim:
            claimName: postgres-pvc
---
apiVersion: v1
kind: Service
metadata:
  name: postgres
spec:
  selector:
    app: postgres
  ports:
    - port: 5432
      targetPort: 5432
```

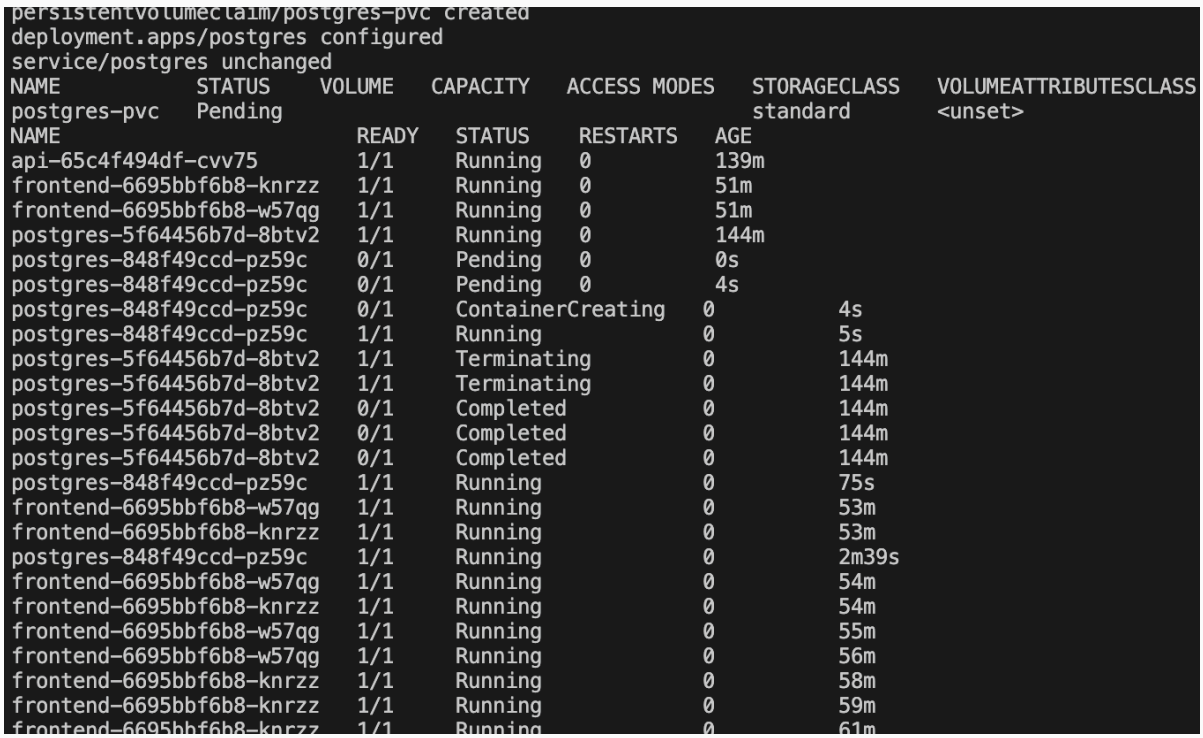
Why the PGDATA environment variable is added

Postgres' own data directory and the volume mount point both default to /var/lib/postgresql/data. Mounting a fresh volume directly there can conflict with a hidden lost+found directory some storage drivers create. Pointing PGDATA at a subfolder (pgdata) inside the mount avoids that conflict entirely.

Step F4 — Apply the updated storage configuration

```
kubectl apply -f 03-postgres-pvc.yaml
kubectl apply -f 03-postgres-with-storage.yaml
kubectl get pvc
kubectl get pods --watch
```

Confirm the PVC shows STATUS Bound, and the postgres pod restarts and reaches 1/1 Running again.



```
persistentvolumeclaim/postgres-pvc created
deployment.apps/postgres configured
service/postgres unchanged
NAME           STATUS    VOLUME   CAPACITY   ACCESS MODES   STORAGECLASS   VOLUMEATTRIBUTESCLASS
postgres-pvc   Pending                                      standard       <unset>
NAME                        READY   STATUS    RESTARTS   AGE
api-65c4f494df-cvv75        1/1     Running   0          139m
frontend-6695bbf6b8-knrzz   1/1     Running   0          51m
frontend-6695bbf6b8-w57qg   1/1     Running   0          51m
postgres-5f64456b7d-8btv2   1/1     Running   0          144m
postgres-848f49ccd-pz59c    0/1     Pending   0          0s
postgres-848f49ccd-pz59c    0/1     Pending   0          4s
postgres-848f49ccd-pz59c    0/1     ContainerCreating   0          4s
postgres-848f49ccd-pz59c    1/1     Running             0          5s
postgres-5f64456b7d-8btv2   1/1     Terminating         0          144m
postgres-5f64456b7d-8btv2   1/1     Terminating         0          144m
postgres-5f64456b7d-8btv2   0/1     Completed           0          144m
postgres-5f64456b7d-8btv2   0/1     Completed           0          144m
postgres-5f64456b7d-8btv2   0/1     Completed           0          144m
postgres-848f49ccd-pz59c    1/1     Running             0          75s
frontend-6695bbf6b8-w57qg   1/1     Running             0          53m
frontend-6695bbf6b8-knrzz   1/1     Running             0          53m
postgres-848f49ccd-pz59c    1/1     Running             0          2m39s
frontend-6695bbf6b8-w57qg   1/1     Running             0          54m
frontend-6695bbf6b8-knrzz   1/1     Running             0          54m
frontend-6695bbf6b8-w57qg   1/1     Running             0          55m
frontend-6695bbf6b8-w57qg   1/1     Running             0          56m
frontend-6695bbf6b8-knrzz   1/1     Running             0          58m
frontend-6695bbf6b8-knrzz   1/1     Running             0          59m
frontend-6695bbf6b8-knrzz   1/1     Running             0          61m
```

Step F5 — Prove data survives a Pod deletion

4. Open a shell into the postgres pod and connect with psql.
5. Create a table and insert one row.
6. Exit, then delete the postgres pod directly.
7. Wait for the replacement pod to reach Running.
8. Connect again and confirm the table and row still exist.

```
kubectl exec -it postgres-5f64456b7d-8btv2 -- psql -U appuser -d appdb

-- inside psql:
CREATE TABLE proof (id SERIAL PRIMARY KEY, note TEXT);
```

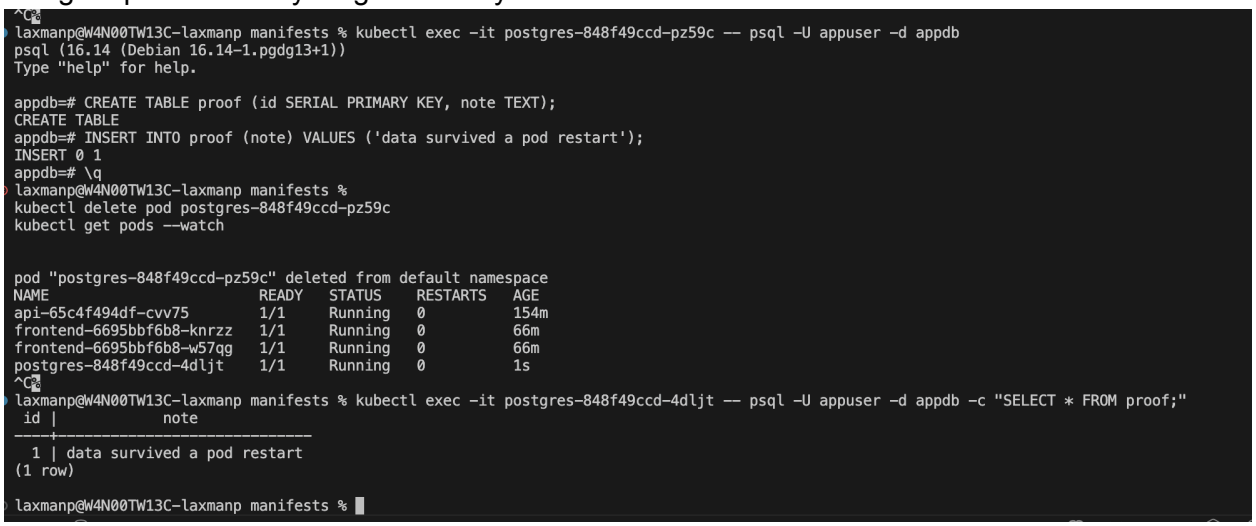
```
INSERT INTO proof (note) VALUES ('data survived a pod restart');
\q

kubectl delete pod postgres-848f49ccd-pz59c
kubectl get pods --watch
```

Once the new pod is Running, reconnect and check:

```
kubectl exec -it postgres-848f49ccd-4dljt -- psql -U appuser -d appdb -c "SELECT *
FROM proof;"
```

The row should still be there. Compare this directly to Lab 1, where deleting the unmounted Postgres pod lost everything -- the only difference now is the PersistentVolumeClaim.



Troubleshooting Reference

Symptom	Likely Cause / Fix
nginx serves a blank page or 404 for any route	The dist output path in the Dockerfile does not match your actual Angular build output. Re-check with dir dist\my-angular-app and adjust the COPY line.
Refreshing on a deep link returns 404	nginx.conf is missing or its try_files directive was not applied. Confirm the file was actually copied into the image.
frontend pod shows ImagePullBackOff or ErrImagePull	If using kind, the image was built on the host but never loaded into the kind node's separate image store. Repeat Step B5.
kubectl get ingress shows no ADDRESS	The Ingress Controller is not yet ready, or did not install correctly. Re-check kubectl get pods -n ingress-nginx.
Browser console shows a MIME-type error for a .js file, page renders blank	A rewrite-target annotation is being applied to the frontend's own Ingress rule, not just the API rule. Confirm the frontend Ingress (frontend-ingress) has no rewrite annotation at all -- only api-ingress should have one.

/api/db-check returns 404 through the Ingress but works via port-forward	The rewrite-target annotation or path regex on api-ingress does not match. Confirm the path is exactly /api(/|$)(.*) and the annotation is /$2.
PVC stuck in Pending	No StorageClass is available to satisfy the claim. On Docker Desktop this is rare, but confirm with kubectl get storageclass that a default class exists.
Postgres pod fails to start after adding the volume	Check kubectl logs <pod-name> for a permissions or lost+found error; confirm PGDATA is set to a subfolder, not the mount root.

Key takeaway

Three separate problems, three separate Kubernetes primitives: Ingress solved "how do two different apps share one entry point," the Ingress Controller solved "who actually implements that routing," and the PersistentVolumeClaim solved "how does data outlive a Pod." None of these existed in Lab 1 by accident -- each was introduced only once its absence became a felt, concrete problem.